Assignment 2: Web Application Modeling (CLO-2)

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1. Content Model (Information Requirements)

The Content Model captures the underlying information structure and dynamic behavior of the web application mapped through the UML-based Web Engineering (UWE) methodology.

1.1 Structural Aspect (Class Diagram Description)

The application relies on four primary logical objects to encapsulate its business domain:

- **User**
  - **Description:** Represents system visitors interacting with the website.
  - **Attributes:** `UserID` , `Name` , `Email` , `Phone` , `SessionToken` .
- **Service**
  - **Description:** Represents the digital offerings provided by IqraSofts.
  - **Attributes:** `ServiceID` , `ServiceName` (e.g., Web Development, Cybersecurity), `Description` , `IconURL` .
- **Project**
  - **Description:** Records portfolio items demonstrating prior digital solutions.
  - **Attributes:** `ProjectID` , `Title` , `ClientName` , `TechStack` , `ImageURL` , `LiveDemoLink` .
- **Inquiry**
  - **Description:** Captures messages from users seeking quotes or consultations.
  - **Attributes:** `InquiryID` , `UserID` (Foreign Key), `MessageContent` , `SubmissionDate` , `Status` .

1.2 Behavioral Aspect (State & Interaction)

The interaction layer bridges the client-side UI and the cloud-hosted intelligence.

- **Frontend Request Generation:** The system detects user interaction (e.g., interacting with the chat widget) and triggers asynchronous JS fetch requests.
- **Backend Processing:** The Python API (managed via `api/index.py` served as a Vercel serverless function) receives HTTP POST requests. It maintains state by parsing the message history payload and authenticates securely against the OpenAI instance.
- **State Transitions:** The system transitions across three visible states: *Idle* (Waiting for input), *Processing* (Typing indicator displayed during API fetch), and *Resolved* (Content successfully injected into the DOM or Error handled gracefully via `500` / `404` catching).

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2. Navigation Model (Navigability)

The Navigation Model defines how users explore the information space, ensuring a seamless user experience prioritizing accessibility.

2.1 Site Map

The application follows a shallow, strictly linked architectural tree to prevent deep nested routing:

- `index.html` : The logical root and landing page.
- `services.html` : Expands upon the core competencies defined in the Service object.
- `projects.html` : Visual representation of the core Project objects.
- `blog.html` : Technical articles and industry updates.
- `team.html` : Profiles of software engineers and leadership.
- `contact.html` : Form handling and geographical map localization.

2.2 Navigation Links and Process Links

- **Navigation Links (Global Menu):** A sticky header exists throughout the session, providing fixed associative links that grant immediate traversal between the six core HTML pages. Mobile dimensions collapse this into a hamburger (toggleable) navigational structure.
- **Process Links:** Certain anchors act as functional triggers rather than mere routing mechanisms. For example, submitting the Contact form or initiating a chat payload triggers the execution of JavaScript processes, transitioning the application's internal data state rather than navigating to a distinct URL.

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3. Presentation Model (User Interface)

The Presentation Model dictates the visual and structural formatting of the data presented to the user, strictly observing responsive design principles.

3.1 UI Hierarchy

The spatial composition of every page follows a uniform trichotomy:

- **Header (Absolute Top):** Contains the official IqraSofts logo, the primary navigation links, and a mobile toggle, housed within a glass-morphic, highly visible container.
- **Main Body (Dynamic Center):** Content clusters mapped via grid systems displaying the current context (e.g., hero banners, specialized grids, contact inputs).
- **Footer (Absolute Bottom):** Stores copyright metadata, secondary navigation, and external social media paths with heavily calibrated bottom padding (`120px`) to prevent spatial conflicts.
- **Floating Action Components (Z-Index Hierarchy):** The WhatsApp direct contact button and the AI Assistant operate independently of the document flow, overlaying the UI asynchronously.

3.2 Interface Behavior and Cross-Device Compatibility

A crucial aspect of the interface's behavior is its environmental awareness, primarily enforced through advanced scripting such as `force_responsive.js` .

- **Dynamic Grid Restructuring:** When the viewport drops below 992px or 768px thresholds, the scripts forcefully redefine `max-width` grids (`.team-grid` , `.services-grid`) by injecting strictly prioritized `!important` inline style blocks.
- **Flex Wrapping Override:** Multi-column flex containers are forcefully converted into single-column flows (`flex-direction: column`) to guarantee absolute legibility without horizontal overflow, specifically on highly constrained devices (e.g., iPhone SE).

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4. Conclusion

By strictly adhering to the UML-based Web Engineering (UWE) methodology, the IqraSofts.com architecture logically isolates content, navigation, and presentation. This disciplined modeling approach has resulted in a robust deployment structure that is both highly scalable—accommodating cloud-hosted Python AI backends—and visually flawless across varying mobile environments. The resulting application offers a secure, intuitive, and modern user experience.